

68retro MP-09 (Revision 3) 6809 Microprocessor Board

The 68retro MP-09 processor board uses the internal crystal oscillator version of the MC6809 or equivalent Hitachi HD6309 in a circuit that is almost identical to that of the MP-02 6802 Microprocessor board. It consists of the MC6809 processor, configurable ROM, RAM and IO circuitry.

The MP-09 board may be used in place of the MP-02 6802 processor board in a SS50-100 bus system and has the following similar features:

HARDWARE FEATURES

- DIP Switch programmable shared 64k memory map for ROM and RAM.
- DIP Switch programmable I/O addressing.
- Expandable, Breadboard Friendly, modified SS50 Bus (SS50-100).
- Up to 56k of RAM.
- Supports a single JEDEC 0.6" Wide 2k/4k/8k/32k/64k x 8 EPROM or 2k/8k/32k/64k x 8 EEPROM.
- Supports a single JEDEC 0.6" Wide 2k/8k/32k/128k/512k x 8 SRAM or a JEDEC 0.3" Wide 64kx8 SRAM.
- Less than half the size of a regular SS50 card.
- 5V only operation.

The 2MHz CMOS HD63B09P and 3MHz HD63C09P can both be used in the MP-09 and have been tested at their respective maximum operating frequencies.

The 50 way SS50 bus connection, re-imagined with a spacing of 0.1", is perfect for expansion and hacking with solderless breadboard and stripboard. Small changes to the bus itself, like making it 5V, make the SS50-100 bus simpler to use.

The unique banked addressing scheme provides a high level of flexibility, allowing existing code to be accommodated without modification at the address where it originally ran.

Similarly, I/O paging allows memory mapped I/O to be set anywhere across the memory map via a DIP switch, overlaying the ROM and RAM so you lose the minimum amount of address space to I/O.

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1. MICROPROCESSOR

Available in three speed grades, 1 MHz (MC6809), 1.5 MHz (MC68A09) and 2 MHz (MC68B09) and available in the P suffix, Plastic package, the L suffix, Ceramic package and the S suffix 'cerdip'.

The Hitachi (now Renesas) CMOS HD6309 came in two speed grades, 2MHz (HD63B09) and 3MHz (HD63C09) the latter of which is highly sought after to replace the NMOS 6809E processor in the Tandy Color Computer, albeit the External clock version HD63C09E.

2. OSCILLATOR

The crystal footprint has been designed for the HC49 case style and has provision for a wire to hold the crystal down against the PCB. A lower profile HC49/U crystal can be used but needs to be installed off the board a little to avoid the possibility of the case of the crystal shorting out the crystal pins by contacting the top layer pads.

In some 'modern' minimal retro systems the microprocessor crystal oscillator is also used as the serial Baud clock. The MP-09 PCA does not make the microprocessor crystal output available on the SS50-100 Bus but the E clock is available for use as a Baud rate clock.

A table of common microprocessor crystals and E clock frequencies is presented below.

MPU Crystal (X1)	Load Capacitors (C3, C4)	E Clock
3.6864 MHz	27 pF	921.6 KHz
3.58 MHz	27 pF	895.0 KHz
4 MHz	27 pF	1 MHz
6 MHz	20 pF	1.5 MHz
7.3728 MHz	18 pF	1.8432 MHz
8 MHz	18 pF	2 MHz
12 MHz	15 pF	3 MHz

Crystal, E clock and loading capacitors.

3. RESET CIRCUIT

The data sheet specifies that Reset be held low for a crystal start up time of 100mS and held low for a minimum of two clock cycles at other times.

The DS1233-5 is the preferred option, holding Reset low for 350mS at power up and when the reset button is pressed. The datasheet also notes that a pullup of minimum 1k and a capacitor between 500pF and 10nF should be connected to the Reset pin when using a push button and additional Reset current is required.

If a RC time constant is chosen then an R (R1) of 3k3 and a C (C2) of 10uF seems to be enough for an adequate Reset.

4. MEMORY DEVICES - ROM

The 28 pin ROM footprint can accept (E)EPROM devices from 2k x8 Bytes to 64k x8 Bytes using five jumpers as presented in the tables below.

EPROM

EPROM	P1A	P1B	P27	P26	P23
2716	NC	NC	NC	1-2	1-2
2732	NC	NC	NC	1-2	2-3
2764	1-2	NC	1-2	NC	2-3
27128	1-2	NC	1-2	2-3	2-3
27256	1-2	NC	2-3	2-3	2-3
27512	2-3	1-2	2-3	2-3	2-3

EEPROM

EEPROM	P1A	P1B	P27	P26	P23
AT28C16	NC	NC	NC	1-2	1-2
AT28C64	NC	NC	1-2	NC	2-3
AT28C256	2-3	2-3	1-2	2-3	2-3
27SF256	1-2	NC	2-3	2-3	2-3
W27C512	2-3	1-2	2-3	2-3	2-3

ROM Strap Settings (NC = no strap connection)

5. MEMORY DEVICES - RAM

The 32 pin RAM footprint can accept SRAM devices from 2k x 8 Bytes to 512k x 8 Bytes using two jumpers as presented in the table below. The MC6802 can only address 64k Bytes so the higher capacity devices can only be addressed up to 64k. The larger devices (> 64k Bytes) are included to help with parts procurement.

SRAM

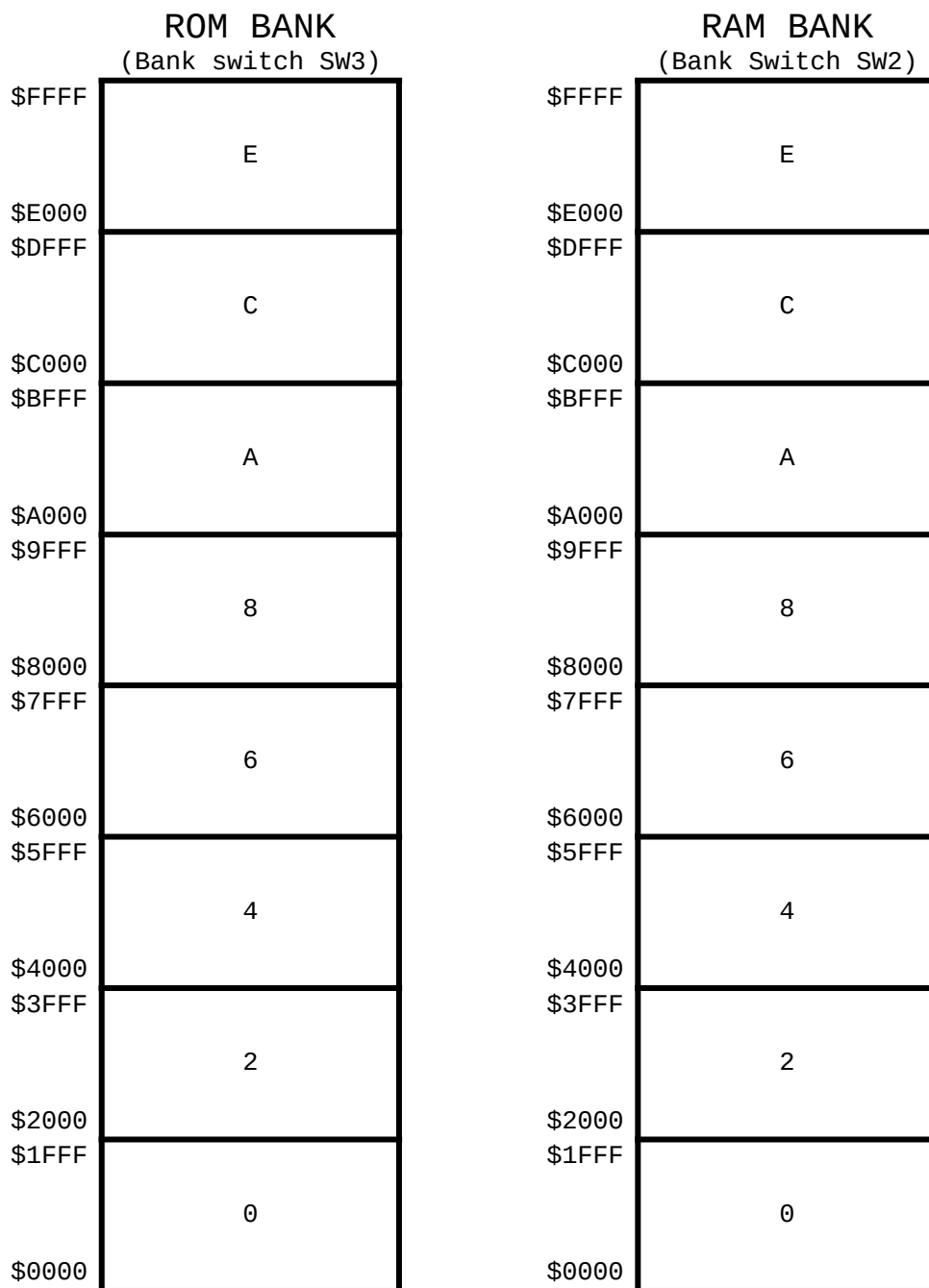
SRAM	P28	P25
6116	1-2	1-2
6264	1-2	2-3
62256	2-3	2-3
628128	2-3	2-3
628512	2-3	2-3

RAM Strap Settings (NC = no strap connection)

6. SS50-100 BUS

The SS50 bus has been changed to make it a little easier to use. The original SS50 bus uses 0.156" pitch Molex connectors; by reducing the pitch of the connector to 0.1" the bus can be used with a variety of commonly available connectors, solderless breadboard and the similar Molex 0.1" connector. The unregulated 7-8V DC power rail has been replaced with a single +5V DC power rail. Compatibility with future processor cards and increased memory addressing has also been catered for with the removal of the baud rate oscillator outputs on the bus. Finally, due to the reduced width of the bus connector, the board size has been reduced by more than 50% compared to a regular SS50 Bus printed circuit board. The MP-09 has a few more signals available on the SS50-100 bus than the MP-02 6802 board,

Figure 1. 68retro MP-09 MEMORY MAP



Memory Mapped I/O

I/O Page (SW4) Upper I/O Address								Lower I/O Address							
A15	A14	A13	A12	A11	A10	A9	A8	A7	A6	A5	A4	A3	A2	A1	A0
\$01 - \$FE*								\$00 - \$FF Decoded on I/O board(s)							

* See Memory Mapped I/O page decoding section.

7. BANK ADDRESS DECODING

The complete 64k memory map is decoded into 8, 8k x8 Byte banks. Both RAM (DIP switch 2) or ROM (DIP switch 3) can be switched in and out of all banks.

The single memory device can be switched to appear in more than one bank and smaller devices, less than 8k, have aliases through the rest of the 8k bank.

Switch the DIP switch to the ON position to enable the RAM or ROM for each 8k bank and make sure you don't enable both RAM and ROM on the same bank!

8. MEMORY MAPPED I/O PAGE DECODING

The Magnitude Comparator (IC8) 74xx688 compares for equality the valid upper byte of the address bus with the I/O PAGE DIP switch setting (SW4). The output active when an address match occurs.

This output is active for all the addresses where the upper byte matches the DIP switch giving a 256 byte I/O address space that can be set to any page in the whole memory map.

The I/O output is provided on the SS-50-100 bus in true and inverted form for additional I/O decoding on other attached boards.

The I/O output also disables the chip selection of both the RAM and the ROM to ensure that neither is enabled during an I/O cycle. In this way only 256 bytes of memory is lost in the whole 64k address space for memory mapped I/O.

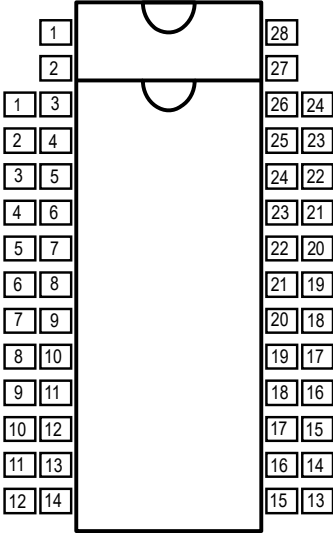
Since the ROM and RAM devices are disabled by the I/O; the top page (\$FF) should not be used as the reset and interrupt vectors are located there.

Figure 2. ROM and RAM Strap settings

	ROM	P1B	P1A	P27	P26	P23		RAM	P28	P25	
EPR0M 2716							6116				1
EEPROM 2816							2k x8				2
											3
EPR0M 2732							6264				
4k x8							8k x8				
EPR0M 2764							62256				
8k x8							32k x8				
EPR0M 27128							628128				
16k x8							128k x8				
EPR0M 27256							628512				
32k x8							256k x8				
(E)EPR0M 27512											
64k x8											
EEPROM 2864											
8k x8											
EEPROM 28256											
32k x8											

Figure 3. EPROM PINOUTS

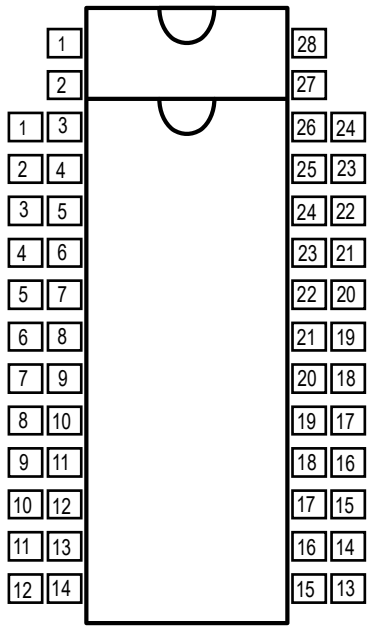
27C512	27C256	27C128	27C64	27C32	27C16
64k x8	32k x8	16k x8	8k x8	4k x8	2k x8
A15	VPP	VPP	VPP		
A12	A12	A12	A12		
A7	A7	A7	A7	A7	A7
A6	A6	A6	A6	A6	A6
A5	A5	A5	A5	A5	A5
A4	A4	A4	A4	A4	A4
A3	A3	A3	A3	A3	A3
A2	A2	A2	A2	A2	A2
A1	A1	A1	A1	A1	A1
A0	A0	A0	A0	A0	A0
D0	D0	D0	D0	D0	D0
D1	D1	D1	D1	D1	D1
D2	D2	D2	D2	D2	D2
GND	GND	GND	GND	GND	GND



27C16	27C32	27C64	27C128	27C256	27C512
2k x8	4k x8	8k x8	16k x8	32k x8	64k x8
		VCC	VCC	VCC	VCC
		PGM	PGM	A14	A14
VCC	VCC	NC	A13	A13	A13
A8	A8	A8	A8	A8	A8
A9	A9	A9	A9	A9	A9
VPP	A11	A11	A11	A11	A11
OE	OE/VPP	OE	OE	OE	OE/VPP
A10	A10	A10	A10	A10	A10
CE	CE	CE	CE	CE	CE
D7	D7	D7	D7	D7	D7
D6	D6	D6	D6	D6	D6
D5	D5	D5	D5	D5	D5
D4	D4	D4	D4	D4	D4
D3	D3	D3	D3	D3	D3

Figure 4. EEPROM PINOUTS

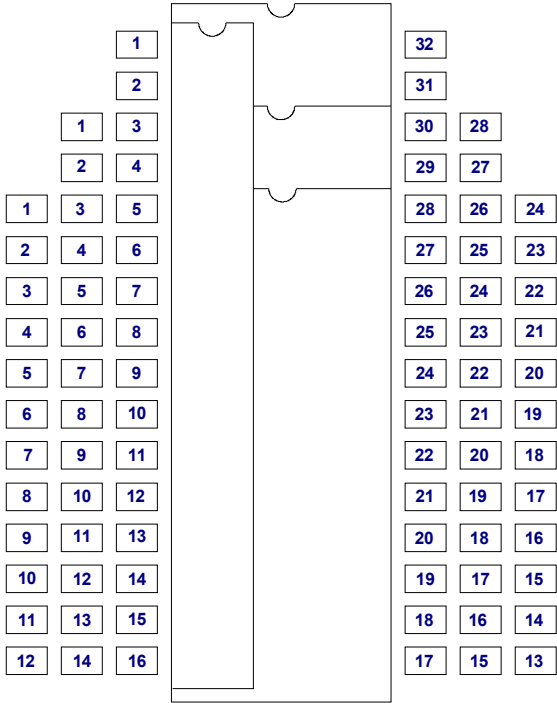
W27C512	27SF256	28C256	28C64	28C16
64k x8	32k x8	32k x8	8k x8	2k x8
A15	VPP	A14	NC	
A12	A12	A12	A12	
A7	A7	A7	A7	A7
A6	A6	A6	A6	A6
A5	A5	A5	A5	A5
A4	A4	A4	A4	A4
A3	A3	A3	A3	A3
A2	A2	A2	A2	A2
A1	A1	A1	A1	A1
A0	A0	A0	A0	A0
D0	D0	D0	D0	D0
D1	D1	D1	D1	D1
D2	D2	D2	D2	D2
GND	GND	GND	GND	GND



28C16	28C64	28C256	27SF256	W27C512
64k x8	8k x8	32k x8	32k x8	64k x8
	VCC	VCC	VCC	VCC
	$\overline{WE}$	$\overline{WE}$	A14	A14
VCC	NC	A13	A13	A13
A8	A8	A8	A8	A8
A9	A9	A9	A9	A9
$\overline{WE}$	A11	A11	A11	A11
$\overline{OE}$	$\overline{OE}$	$\overline{OE}$	$\overline{OE}/VPP$	$\overline{OE}/VPP$
A10	A10	A10	A10	A10
$\overline{CE}$	$\overline{CE}$	$\overline{CE}$	$\overline{CE}$	$\overline{CE}$
D7	D7	D7	D7	D7
D6	D6	D6	D6	D6
D5	D5	D5	D5	D5
D4	D4	D4	D4	D4
D3	D3	D3	D3	D3

Figure 5. SRAM PINOUTS

628512	628128	62256	6264	6116
512k x8	128k x8	32k x8	8k x8	2k x8
A18	NC			
A16	A16			
A14	A14	A14	NC	
A12	A12	A12	A12	
A7	A7	A7	A7	A7
A6	A6	A6	A6	A6
A5	A5	A5	A5	A5
A4	A4	A4	A4	A4
A3	A3	A3	A3	A3
A2	A2	A2	A2	A2
A1	A1	A1	A1	A1
A0	A0	A0	A0	A0
D0	D0	D0	D0	D0
D1	D1	D1	D1	D1
D2	D2	D2	D2	D2
GND	GND	GND	GND	GND



6116	6264	62256	628128	628512
2k x8	8k x8	32k x8	128k x8	512k x8
			VCC	VCC
			A15	A15
		VCC	VCC	$\overline{CS2}$ A17
		$\overline{WE}$	$\overline{WE}$	$\overline{WE}$
VCC	$\overline{CS2}$	A13	A13	A13
A8	A8	A8	A8	A8
A9	A9	A9	A9	A9
$\overline{WE}$	A11	A11	A11	A11
$\overline{OE}$	$\overline{OE}$	$\overline{OE}$	$\overline{OE}$	$\overline{OE}$
A10	A10	A10	A10	A10
$\overline{CE}$	$\overline{CS1}$	$\overline{CE}$	$\overline{CS1}$	$\overline{CE}$
D7	D7	D7	D7	D7
D6	D6	D6	D6	D6
D5	D5	D5	D5	D5
D4	D4	D4	D4	D4
D3	D3	D3	D3	D3